Regular polyhedra with an odd kernel of index four

The human_compression project

Canonical Topic F: application

Abstract

For every odd prime p, we recover the structural and abelian-kernel classification of rank-three string C-groups of order $4p^m$ and give their nonabelian family in cyclotomic coordinates. The construction is identified generator by generator with the original presentation. The argument separates a two-character classification from a ring calculation valid for all parameters. A literal marking in the specified primary edition is corrected explicitly.

1 The structural theorem

We use the twelve-page preprint arXiv:2407.10388v1 by Hou, Feng, Leemans and Qu [2]. We recover its Theorems 1.1–1.3 and the explicit construction in Theorem 5.1, with the correction stated below. The results remain attributed to those authors. This edition reorganizes the accepted proofs through the common paper [1]; no previous combined manuscript is an additional proof premise.

We use the conventions of that paper: marked generators are distinct nonidentity involutions, the endpoints commute, and the adjacent orders are at least three. A string C-group satisfies the full subset intersection property. Its type is $\{k_1, k_2\}$, and it is tight when $|G| = 2k_1k_2$. Conjugation is $x^g = g^{-1}xg$.

Theorem 1 (The odd kernel)**.** *Let p be an odd prime and $m \geq 2$. A nondegenerate rank-three string C-group $(G; r_0, r_1, r_2)$ of order $4p^m$ has a normal Sylow p-subgroup P, and*

$$G = P \rtimes K, \qquad K = \langle r_0, r_2 \rangle \cong C_2 \times C_2.$$

Both type parameters are divisible by p, at least one is divisible by $2p$, and P needs exactly two generators.

Proof. The number n_p of Sylow p-subgroups divides four and is congruent to one modulo p. The only case other than $n_p = 1$ is $p = 3$, $n_p = 4$. Assume this case occurs. Conjugation on the four Sylow subgroups gives a homomorphism $G \longrightarrow S_4$ with kernel $N \leq N_G(P) = P$, since the normalizer has index four. The image has order $4 \cdot 3^j$ dividing 24, so $j = 0$ or 1. If $j = 0$, then $N = P$ and P is normal, a contradiction. If $j = 1$, the image is the index-two subgroup A_4 of S_4. Its involutions all lie in the normal Klein four subgroup. They cannot generate A_4, whereas an image of an involution-generated group is generated by involutions and identities. This is also a contradiction. Thus P is normal.

The endpoint subgroup K has order four, since its two generators are distinct commuting involutions. It intersects the odd group P trivially, and $|PK| = |G|$, giving the stated semidirect product. Each adjacent dihedral group has order $2k_i$, so $k_i \mid 2p^m$. Nondegeneracy implies $p \mid k_i$. If both k_i were odd, their images in the exponent-two quotient $G/P = K$ would force $r_0P = r_1P = r_2P$, contrary to the distinct images of r_0, r_2. Thus one label is even.

The inverted-orbit lemma of [1] gives two generators for P. If P were cyclic, the nontrivial p-parts of both adjacent rotation groups would contain its unique subgroup of order p. That subgroup would lie in $G_{\{0,1\}} \cap G_{\{1,2\}} = \langle r_1 \rangle$, which has order two. This contradiction proves that two generators are necessary. □

2 Abelian kernels and tight markings

Theorem 2 (The abelian classification)**.** *Under the hypotheses of Theorem 1, the kernel P is abelian if and only if the marking is tight. Up to duality, all such markings are the following, one for each $1 \le a \le b$, $a + b = m$:*

$$P = \langle x \rangle \times \langle y \rangle, \qquad |x| = p^b, \quad |y| = p^a,$$
$$x^{r_0} = x, \quad y^{r_0} = y^{-1}, \qquad x^{r_2} = x^{-1}, \quad y^{r_2} = y^{-1}, \qquad r_1 = xyr_2.$$

The type is $\{2p^a, p^b\}$. For this type the marked group is unique.

Proof. First suppose P is abelian. Write $r_1 = ut$ with $u \in P$ and $1 \ne t \in K$. The two-character lemma of [1] shows that P is the direct product of the two nonzero cyclic character components on which t acts as -1. Choose their projected generators x, y so that $u = xy$.

If $t = r_0 r_2$, the two characters on (r_0, r_2) are $(+,-)$ and $(-,+)$. A direct product calculation gives adjacent orders $2|y|$ and $2|x|$. The rank-three product bound of [1] would give $8|x||y| \le |G| = 4|x||y|$, which is impossible. Thus t is an endpoint; by duality take $t = r_2$. The two characters are now $(+,-)$ and $(-,-)$, exactly as in the displayed model. For $|x| = p^b$ and $|y| = p^a$ the adjacent orders are

$$|r_0 r_1| = 2p^a, \qquad |r_1 r_2| = p^{\max(a,b)}.$$

The same bound implies $a + \max(a, b) \le a + b$, hence $a \le b$. It is then an equality; the marking is tight.

Conversely, construct the displayed semidirect product for $1 \le a \le b$. The two specified automorphisms generate a Klein four complement, and r_1 is an involution because r_2 inverts xy. The three marked elements are distinct, the endpoints commute, and the adjacent orders are $2p^a, p^b$. The elements $u = xy$ and $u^{r_0} = xy^{-1}$ yield x^2, y^2; squaring is invertible on both cyclic odd-order factors. Hence the marking generates G.

Put $H = \langle r_0, r_1 \rangle$ and $J = \langle r_1, r_2 \rangle$. The image of H in K has order four. Its rotation square generates $\langle y \rangle$, and $|H| = 4p^a$, so $H \cap P = \langle y \rangle$. Similarly $J \cap P = \langle xy \rangle$ and its image in K is $\langle r_2 \rangle$. These two odd subgroups meet trivially: if $x^j = 1$ then $p^b \mid j$, which also gives $y^j = 1$ since $a \le b$. The map $H \cap J \longrightarrow K$ is consequently injective, with image in $\langle r_2 \rangle$. It contains r_1, so $H \cap J = \langle r_1 \rangle$. The rank-three test proves the full intersection property. The cyclic factor orders, their projected generators and the two characters determine the marked group, giving uniqueness among abelian-kernel models.

It remains to show that tightness implies an abelian kernel. By Theorem 1, each type parameter is a power of p or twice a power of p. The equality $|G| = 2k_1k_2 = 4p^m$ forces exactly one parameter to be even, and the exponents sum to m. We now retain the precise odd/even clause of Cunningham–Pellicer's classification [3, Theorem 1.1(b),(c)]: apart from its exceptional even parameter four, a tight regular polyhedron with one odd and one even type parameter exists only when the even parameter divides twice the odd parameter, and the marked polyhedron is unique for such a type. Here the even parameter is $2p^a$, never four. Thus $a \le b$, and uniqueness identifies the marking with the abelian model just constructed. The use of that theorem includes the retained regular-polyhedron/string-C-group correspondence. This completes the proof. □

The literal formula in the primary edition. The display in [2, Theorem 1.2, physical p. 2] gives $x^{r_0} = x$ and $y^{r_0} = y^{-1}$ but writes $r_1 = xyr_0$. Since x, y commute, this element has square $x^2 \ne 1$. It is not an involution for the asserted positive odd-prime exponent. The body of the same edition, Section 4 on physical page 6, uses xyr_2 in the corresponding character case. Theorem 2 uses that formula and assigns the larger exponent to x. This is a correction to the specified preprint display, not a recovery of its impossible literal formula or an assertion about a later journal edition.

3 A nonabelian kernel for every parameter

Theorem 3 (The cyclotomic polyhedra)**.** *For every odd prime p and every $m \geq 3$, there is a string C-group of order $4p^m$ and type $\{p, 2p\}$ with nonabelian Sylow p-subgroup. More precisely, let*

$$O = \mathbb{Z}[\zeta_p], \quad \pi = \zeta_p - 1, \quad A = O/\pi^{m-1}O, \quad P = A \rtimes \langle\beta\rangle, \qquad |\beta| = p, \quad v^\beta = \zeta_p v.$$

Define automorphisms by

$$\sigma(v) = \overline{v}, \quad \sigma(\beta) = \beta^{-1}, \qquad \tau(v) = -v, \quad \tau(\beta) = \beta.$$

Then $G = P \rtimes \langle\sigma, \tau\rangle$ has the required marking

$$r_0 = s\tau\sigma, \qquad r_1 = \beta\tau\sigma, \qquad r_2 = \sigma,$$

where s denotes the additive class of the ring element 1. It is not the identity of the additive group A.

Proof. Set $n = m - 1 \geq 2$ and write $\zeta = \zeta_p$. To fix the action convention, use pairs $(v, k) \in A \times \mathbb{Z}/p\mathbb{Z}$ with

$$(v, k)(w, l) = (v + \zeta^{-k}w, k + l). \tag{1}$$

Then $\beta = (0, 1)$ has $v^\beta = \zeta v$. The cyclotomic coordinates of [1] give $|P| = p^{n+1} = p^m$. The action by ζ is nontrivial for $n \geq 2$, so P is nonabelian. Every element outside A has order exactly p: for $k \neq 0$ its pth power has first coordinate $\sum_{i=0}^{p-1} \zeta^{-ik}v = 0$, and its image in P/A has order p.

On pairs the proposed maps are $\sigma(v, k) = (\overline{v}, -k)$ and $\tau(v, k) = (-v, k)$. They preserve (1), since conjugation sends ζ^{-k} to ζ^k. They are commuting involutory automorphisms. Moreover, σ inverts the nontrivial element β, whereas τ fixes it; and τ is nontrivial on the nonzero odd-order group A. Thus they are independent and generate a Klein four group. In particular $|G| = 4p^m$.

Put $h = \tau\sigma$. In multiplicative group notation $s^h = s^{-1}$ and $\beta^h = \beta^{-1}$. The elementary reflection identities of [1] make $r_0 = sh$, $r_1 = \beta h$ and $r_2 = \sigma$ involutions. Directly,

$$r_0r_2 = s\tau = r_2r_0, \qquad r_0r_1 = s\beta^{-1}, \qquad r_1r_2 = \beta\tau.$$

The middle product is outside A and has order p. The last has order $2p$, since its two factors commute and have coprime orders. The marked generators are nonidentity and distinct; the first two project to $\tau\sigma$ in the complement, the third to σ, and the first two are distinct because $s \neq \beta$.

They generate the whole group. Indeed, $(r_1r_2)^p = \tau$ yields τ, then β; the generator r_2 gives σ, and r_0 gives s. The β-conjugates of s are $1, \zeta, \ldots, \zeta^{p-1}$, whose classes additively generate A.

Finally $H = \langle r_0, r_1\rangle$ and $J = \langle r_1, r_2\rangle$ have orders $2p$ and $4p$. The subgroup $C = \langle r_1\rangle$ has prime index p in H. If $H \leq J$, all three generators would lie in J, giving $G = J$ and contradicting $4p^m > 4p$. The prime-index lemma of [1] therefore gives $H \cap J = C$. Its rank-three test proves the full intersection property. □

4 The exact presentation and its automorphisms

Theorem 4 (The source-generator identification)**.** *Write*

$$m - 1 = (e - 1)(p - 1) + r, \qquad e \geq 1, \quad 1 \leq r \leq p - 1,$$

and put $d_i = p^e$ for $i \leq r$ and $d_i = p^{e-1}$ for $i > r$. The Sylow subgroup in Theorem 3 has the exact presentation

$$\begin{aligned}\langle s_1, \ldots, s_{p-1}, \beta \mid s_i^{d_i} = 1, \quad \beta^p = 1, \quad [s_i, s_j] = 1,\\ [s_i, \beta] = s_{i+1}\ (1 \leq i < p - 1),\\ [s_{p-1}, \beta] = \prod_{i=1}^{p-1} s_i^{-\binom{p}{i}}\rangle.\end{aligned}$$

The power and commutativity relators range over $1 \le i, j \le p-1$. *The generator* s_i *maps to* π^{i-1} *in* A *and has exactly order* d_i. *Under this isomorphism,* σ *fixes* s_1 *and inverts* β, *while* τ *inverts* s_1 *and fixes* β. *Thus the group, automorphisms and marking are precisely those of [2, Section 5 and Theorem 5.1].*

Proof. The coordinate theorem of [1] gives the exact additive orders and commutativity of the proposed s_i. Our commutator convention gives $[s_i, \beta] = \pi s_i = s_{i+1}$ for $i < p-1$. The cyclotomic equation

$$p + \binom{p}{2}\pi + \cdots + \binom{p}{p-1}\pi^{p-2} + \pi^{p-1} = 0$$

gives the final relation. Thus all literal source relators hold.

Conversely let $\widehat{P}$ be the group defined by the displayed presentation. Its subgroup $S = \langle s_1, \dots, s_{p-1} \rangle$ is finite abelian, with order at most $\prod_i d_i = p^{er+(e-1)(p-1-r)} = p^{m-1}$. The conjugation relators give $S^\beta \le S$; since conjugation is injective and S is finite, this is equality. Hence S is normal and $\widehat{P}/S$ is cyclic of order at most p. Consequently $|\widehat{P}| \le p^m$. The homomorphism to the constructed group P is surjective: the displayed basis generates A, and β generates the quotient. Its image has order p^m, so the homomorphism is an isomorphism. This proves the generator-by-generator assertion, including modulus-one generators.

The formulas on s_1, β are the definitions of σ, τ. These elements generate P, because successive commutators of s_1 with β give every s_i. The formulas therefore identify the automorphisms, not only their effects on a proper subgroup. □

The auxiliary binomial formulas. Define $s_p = [s_{p-1}, \beta]$ and $s_{p+1} = [s_p, \beta]$. They are the additive classes π^{p-1}, π^p; they are not additional independent generators. The common conjugation identities give, in multiplicative notation, for $1 \le k \le p$,

$$s_k^{\beta^{1-k}} = \prod_{i=k}^{p+1} s_i^{\binom{p+1-k}{i-k}},$$

$$s_k^\sigma = \begin{cases} s_k^{\beta^{1-k}}, & k \text{ odd}, \\ (s_k^{\beta^{1-k}})^{-1}, & k \text{ even}. \end{cases}$$

These are exactly Lemma 5.2(1),(2) of [2]. The verified automorphisms in Theorem 3 give Lemma 5.3 directly. The source's coefficient-table calculation is therefore unnecessary for this proof: the full relations are preserved by actual automorphisms of the ring model and the exact presentation isomorphism transports them back.

Nilpotency class and the Hughes hypothesis. The group A is abelian, P/A is cyclic, and $[A, \beta] = \pi A$. Induction gives

$$\gamma_j(P) = \pi^{j-1}A \qquad (j \ge 2).$$

The successive lattice indices show that these groups are nontrivial through $j = m-1$ and trivial at $j = m$. Thus P has nilpotency class $m-1$, the maximal-class property stated in the source. The motivating condition $A = H_p(P)$ in its Section 5 is not a condition needed for the constructed presentation. In fact, if $m - 1 \le p - 1$, the coordinate theorem makes A elementary abelian. All elements outside A have order p as well, so every element of P has order dividing p. The Hughes subgroup, generated by elements of order different from p, is then trivial. No general classification of Hughes groups or groups of maximal class is used in any of our arguments.

5 Source scope and verification

Theorem 1 is the structural result [2, Theorem 1.1]. Theorem 2 recovers its Theorem 1.2 with the explicit marking correction. Theorem 3 and the exact presentation recover Theorems 1.3 and 5.1. The auxiliary rank-three criterion, character decomposition, binomial identities, automorphisms and maximal-class assertion are included. The original small-group numerical census, survey discussion and general Hughes classification are outside this principal-results contract.

The mathematical results retain their original attribution. The common inverted-orbit and two-character arguments reorganize the source proofs; the cyclotomic module supplies a uniform replacement for the construction and its presentation checks. We make no novelty claim for the underlying cyclotomic model. In the tightness converse we retain exactly [3, Theorem 1.1(b),(c)] in the stated odd/even range. That primary theorem and the polyhedron correspondence were read in the earlier accepted audit, but its full PDF is not retained locally. There is no local byte binding or known complete-text cost for that external proof. It is not silently replaced by finite computations.

The complete accompanying validation appendix retains three programs: the mixed-modulus model generator, the finite marked-group checker, and the independently authored literal-presentation checker. Together they test seven cyclotomic and four abelian marked models and five literal source presentations. Every tested marking is checked against all 64 subset-intersection identities. The programs also reject the incorrect literal marking, a corrupted cyclotomic relation and altered source generator orders; a composite-index example shows that the prime-index hypothesis is necessary. These are bounded checks of the uniform proofs above, whose conclusions do not depend on a finite enumeration or an unreported program output. The three complete programs remain visible and chargeable as validation; both complete historical operational drivers are inventoried separately.

The five reused Lean declarations verify the two prime-index interfaces and three group identities. Sylow normality, the two-character theorem, cyclotomic lattice calculations, all-parameter construction and source-presentation isomorphism remain written proofs. The retained tight-polyhedron classification and classical correspondence are also outside that formal scope. This edition neither solves a new open question nor changes the previously admitted source count. It asserts no compression ratio; the common paper, complete application, validation programs and external inputs have separate, explicit boundaries.

References

[1] The human_compression project, *Odd kernels and cyclotomic modules*, accompanying common paper.

[2] D.-D. Hou, Y.-Q. Feng, D. Leemans and H.-P. Qu, *String C-groups of order* $4p^m$, arXiv:2407.10388v1, 15 July 2024, 12 pp.

[3] G. Cunningham and D. Pellicer, *Classification of tight regular polyhedra*, arXiv:1408.6703v2, 10 November 2015, Theorem 1.1(b),(c); published in *Journal of Algebraic Combinatorics* 43 (2016), 665–691.